

Western Norway
University of
Applied Sciences

The Primal-Dual Method

Chapter 7

Sven-Olai Høyland

Primal-Dual method

- LPs are efficiently solvable and algorithms to solve them are quick in practice



Special purpose algorithms are often much faster

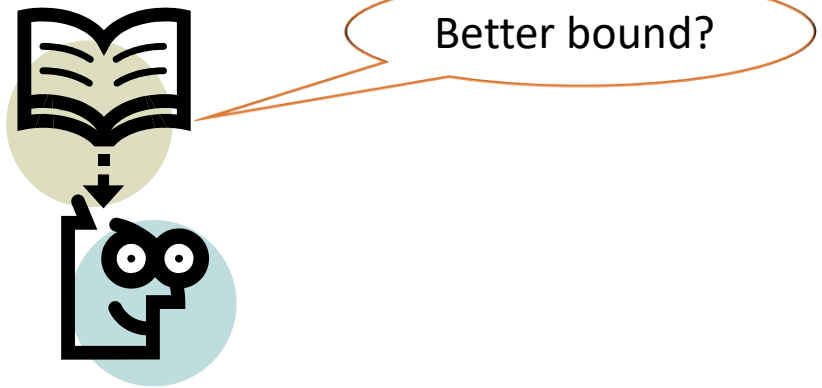
Dual Problem

- Theory
  - Appendix A
- First some motivation, then example

Lower bound on objective function

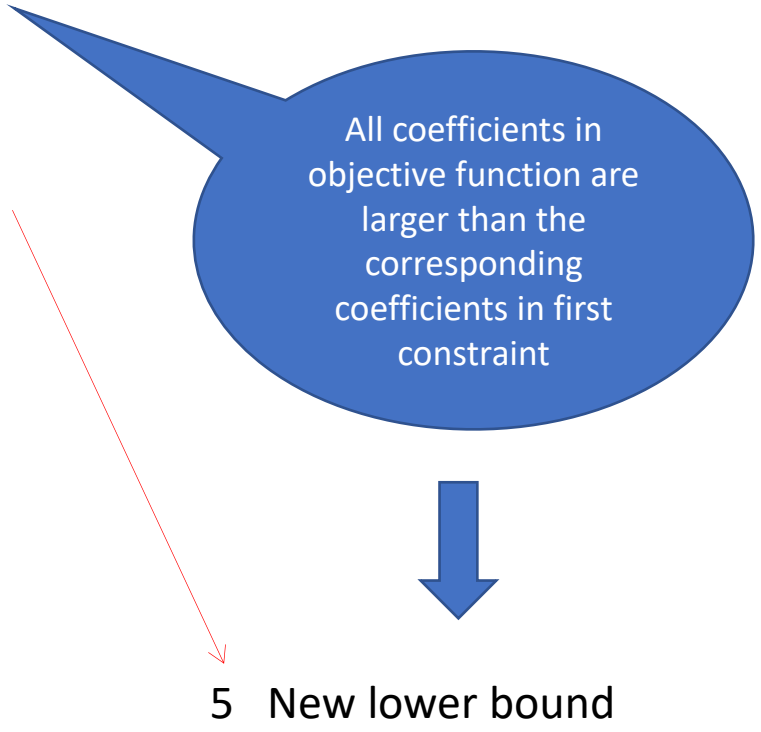
- Minimize $6x_1 + 4x_2 + 2x_3$
- Subject to
  - $4x_1 + 2x_2 + x_3 \geq 5$
  - $2x_1 + x_2 \geq 3$
  - $x_2 + x_3 \geq 4$
  - $x_i \geq 0, \quad i = 1,2,3$

Obviously optimal solution must be at least 0 since all $x_i \geq 0$



Lower bound on objective function

- Minimize $6x_1 + 4x_2 + 2x_3$
- Subject to
  - $4x_1 + 2x_2 + x_3 \geq 5$
  - $2x_1 + x_2 \geq 3$
  - $x_2 + x_3 \geq 4$
  - $x_i \geq 0, \quad i = 1,2,3$



Lower bound on objective function

Adding all three constraints

$$6x_1 + 4x_2 + 2x_3 \geq 12$$

Objective function on left side

New lower bound: 12

Lower bound on objective function

- Were lucky in example! ☺
- But we could do it more systematic
  - Want to find the largest lower bound
  - Using a linear combination of the constraints
  - Formally, next slide

Lower bound on objective function

- Minimize $6x_1 + 4x_2 + 2x_3$
- Subject to
  - y_1 | $4x_1 + 2x_2 + x_3 \geq 5$
  - y_2 | $2x_1 + x_2 \geq 3$
  - y_3 | $x_2 + x_3 \geq 4$
  - $x_i \geq 0, \quad i = 1,2,3$
- Multiplying the constraints with y_i and collecting similar element, we get a new LP problem, next slide

Lower bound on objective function

- Maximize $5y_1 + 3y_2 + 4y_3$
- Subject to
  - $4y_1 + y_2 \leq 6$
  - $2y_1 + y_2 + y_3 \leq 4$
  - $y_1 + y_3 \leq 2$
  - $y_i \geq 0, \quad i = 1,2,3$

Some observations

- LP
  - n variables
  - m constraints
- Dual
  - m variables
  - n constraints
  - Right hand sides in the LP’s constraints becomes coefficients in the Dual’s objective function
  - Coefficients in the LP’s objective function becomes right hand sides in the Dual’s constraints

Important properties

- Any feasible solution to the dual LP has a value not greater than any feasible solution to the primal LP (primal LP is a minimization problem)
  - $\sum_{i=1}^{m} b_i y_i \leq \sum_{j=1}^{n} w_j x_j$ (weak duality property)
- Strong duality property of linear programs
  - As long there exist feasible solution to both the primal and the dual LP, their optimal value are equal
  - Example using Set Cover problem: If x^* is an optimal solution to the set cover LP relaxation and y^* is an optimal solution to the dual LP, then
  - $\sum_{j=1}^{n} w_j x_j^* = \sum_{i=1}^{m} b_i y_i^*$

Set cover problem

- Ground set of elements $E = \{e_1, e_2, \ldots, e_n\}$
- Subsets of the elements $S_1, S_2, \ldots, S_m$ where $S_j \subseteq E$
- Each subset has a nonnegative weight $w_j \geq 0$

- Goal: Find a minimum weight collection of subsets that covers all of E. That is find a subset I of $\{1, 2, \ldots, m\}$ that minimizes $\sum_{j \in I} w_j$ subject to $\bigcup_{j \in I} S_j = E$

Set cover as ILP

minimize $\sum_{j=1}^{m} w_j x_j$

subject to $\sum_{j:e_i \in S_j} x_j \geq 1, \quad i = 1, \ldots, n$

$x_j \in \{0,1\}, \quad \mathrm{j} = 1, \ldots, m$

Example

- $S = \{1,2,3,4\}$
- Subsets
  - $S_1 = \{1,2\}, w_1 = 2$
  - $S_2 = \{1,3\}, w_2 = 1$
  - $S_3 = \{2,4\}, w_3 = 2$
  - $S_4 = \{3,4\}, w_4 = 3$
- minimize $2x_1 + 1x_2 + 2x_3 + 3x_4$
- subject to «all elements must be present in at least one of the subsets chosen»
- $x_j \in \{0,1\}, j = 1, \dots, 4$

Set cover

ILP

minimize $\sum_{j=1}^{m} w_j x_j$

subject to $\sum_{j:e_i \in S_j} x_j \geq 1, \quad i = 1, \ldots, n$

$x_j \in \{0,1\}, \quad \mathrm{j} = 1, \ldots, m$

LP relaxation

minimize $\sum_{j=1}^{m} w_j x_j$

subject to $\sum_{j:e_i \in S_j} x_j \geq 1, \quad i = 1, \ldots, n$

$x_j \geq 0, \quad \mathrm{j} = 1, \ldots, m$

Dual formulation

maximize $\sum_{j=1}^{n} y_i$

subject to $\sum_{i:e_i \in S_j} y_i \leq w_j, \quad i = 1, \ldots, m$

$y_i \geq 0, \qquad \mathrm{i} = 1, \ldots, n$

Example

- $S = \{1,2,3,4\}$
- Subsets
  - $S_1 = \{1,2\}, w_1 = 2$
  - $S_2 = \{1,3\}, w_2 = 1$
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- minimize $2x_1 + 1x_2 + 2x_3 + 3x_4$
- subject to «all elements must be present in at least one of the subsets chosen»
- $x_j \in \{0,1\}, j = 1, \dots, 4$

- Dual?

Dual formulation

- Maximize $y_1 + y_2 + y_3 + y_4$
- Subject to
  - $y_1 + y_2 \leq 2$
  - $y_1 + y_3 \leq 1$
  - $y_2 + y_4 \leq 2$
  - $y_3 + y_4 \leq 3$
  - $y_i \geq 0, \quad i = 1,2,3,4$

Primal-dual algorithm

$y \leftarrow 0$
$I \leftarrow \emptyset$

while there exists $e_i \notin \bigcup_{j \in I} S_j$ **do**

Increase the dual variable y_i until there is some ℓ with $e_i \in S_\ell$ such that $\sum_{j: e_j \in S_\ell} y_j = w_\ell$

$I \leftarrow I \cup \{\ell\}$

Example

$S_1 = \{1,2\}, w_1 = 2$
$S_2 = \{1,3\}, w_2 = 1$
$S_3 = \{2,4\}, w_3 = 2$
$S_4 = \{3,4\}, w_4 = 3$

$y_1 + y_2 \leq 2$
$y_1 + y_3 \leq 1$
$y_2 + y_4 \leq 2$
$y_3 + y_4 \leq 3$

$y \leftarrow 0$
$I \leftarrow \emptyset$

while there exists $e_i \notin \bigcup_{j \in I} S_j$ **do**

Increase the dual variable y_i until there is some ℓ with $e_i \in S_\ell$ such that $\sum_{j: e_j \in S_\ell} y_j = w_\ell$

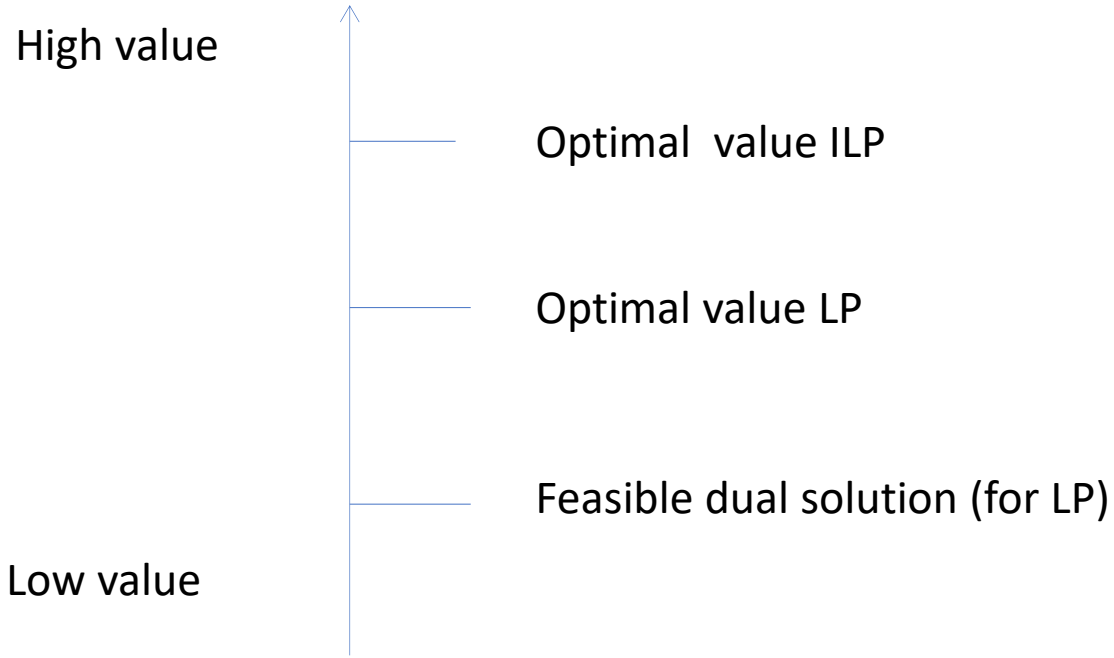
$I \leftarrow I \cup \{\ell\}$

One solution

- y=(0,0,0,0), I = ∅
- Choose y_1, Can be increased to 1 (inequality 2)
  y =(1,0,0,0), I = {2}
- Choose y_2, Can be increased to 1 (inequality 1)
  y =(1,1,0,0), I = {1,2}
- Choose y_4, Can be increased to 1 (inequality 3)
  y =(1,1,0,1), I = {1,2,3}
- Done
- Start by choosing y_4 first

Set Cover Problem

- For all minimization problems. May be equality



For maximization problems it is the other way around

- Theorem 1.9. The Primal-dual algorithm is an f-approximation algorithm for the set cover problem

- f is the maximum number of sets any element is included

Integrality gap

- The integrality gap of an integer program formulation is the worst-case ratio over all instances of the problem of the optimal value of the integer program to the optimal value of the linear program
- Performance guarantee of a primal-dual algorithm provides upper bound on integrality gap
- Integrality gap gives lower bound on the performance guarantee

Not in curriculum

- Chapter 7.2 & 7.3